

Recommendation Systems for Personalized Content Discovery

Netflix Prize Dataset | Matrix Factorization & Collaborative Filtering

100M+ Ratings | **480K Users** | **17,770 Movies** | June 2025

Problem Overview & Dataset

The Challenge

Task: Predict ratings + rank unseen movies

Input: User's historical star ratings (1–5)

Output: Personalised Top-10 movie list

RMSE: Measures rating prediction accuracy

MAP@10: Measures ranking quality (rel. ≥ 4)

Split: Time-based 80/20 per user

Dataset at a Glance

100,480,507

Total Ratings

41,335

Active Users (subset)

17,770

Movies

94.6%

Sparsity

3.60

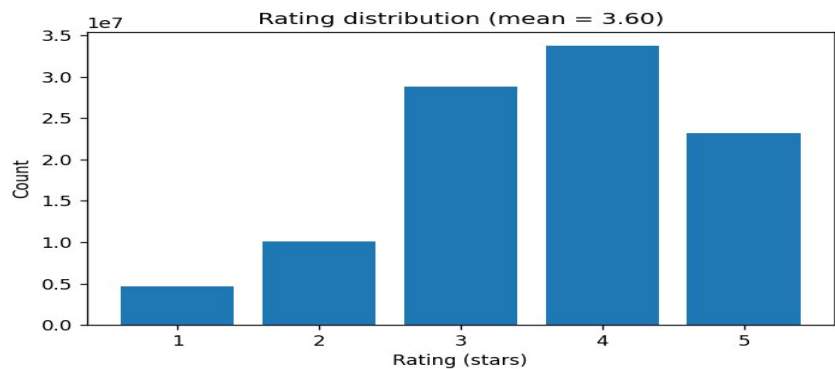
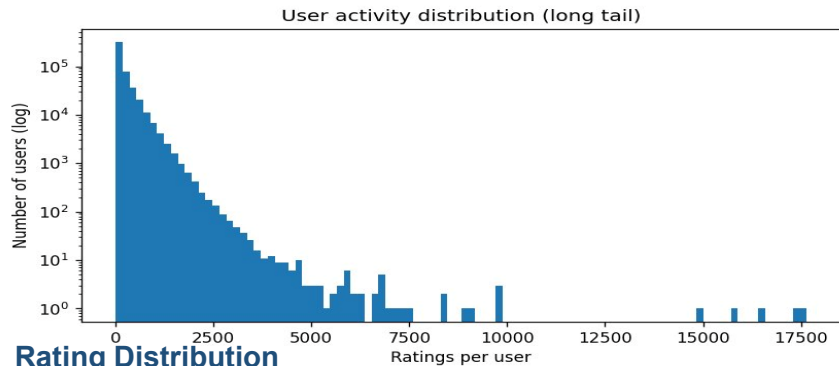
Mean Rating

89.2%

Ratings in top 20% movies

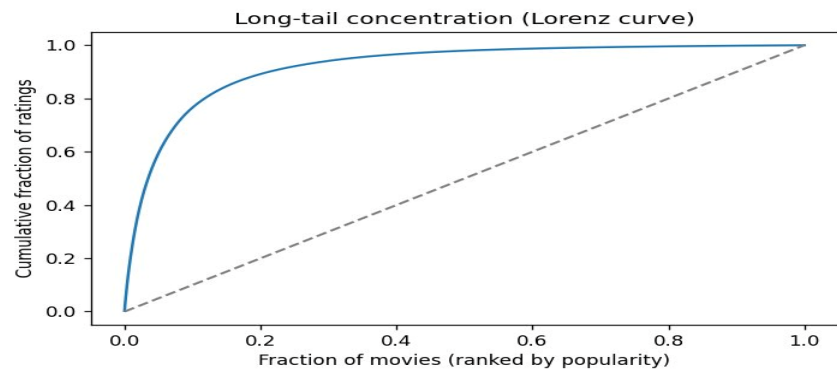
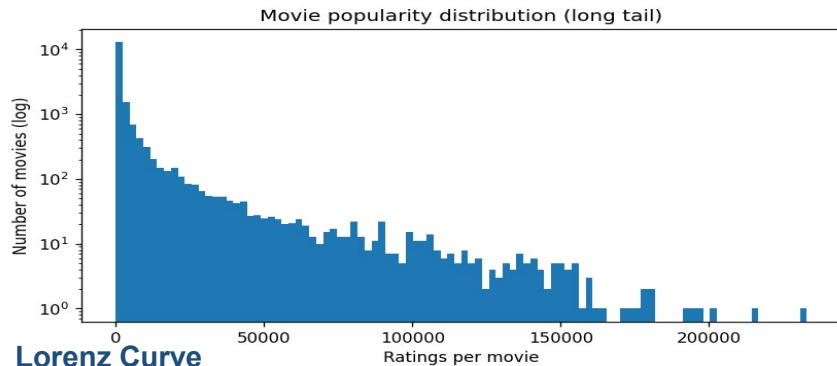
EDA Key Insights

User Activity — Long Tail



Mean 3.60 — users skew positive

Movie Popularity — Long Tail



Top 20% movies hold 89.2% of ratings

Our Approach — Three-Level Stack

Level 1: Baselines

- Global Mean — predict 3.60 for all users and movies
- User Mean — predict each user's own average rating
- Item Mean — predict each movie's average rating

Level 2: Item-Based CF

- Adjusted cosine similarity between movies
- Top-100 neighbours per movie stored on GPU
- Weighted prediction from user's rated neighbours

Level 3: Funk-SVD (Primary)

- $\hat{r} = \mu + b_u + b_i + p_u \cdot q_i$
- 50 latent factors, Adam optimiser, early stopping
- Trained on 32M ratings with GPU acceleration

Funk-SVD — Model Details

$$\hat{r}(u,i) = \mu + b_u + b_i + p_u \cdot q_i$$

μ — Global mean rating (3.60)

b_u — User bias — captures how lenient/strict the user is

b_i — Item bias — captures the movie's intrinsic quality

$p_u \cdot q_i$ — Dot product of 50-dim latent factor vectors

Key Hyperparameters

50

Factors

0.01

LR

0.02

Reg

15

Epochs

100K

Batch

Evaluation Setup

Train / Test Split Strategy

TRAIN (earliest 80% per user)

32,016,146 ratings

TEST 20%

7,983,331 ratings

Evaluation Metrics

RMSE

Root Mean Squared Error — measures rating prediction accuracy.

$$\text{RMSE} = \sqrt{\text{mean}((r - \hat{r})^2)}$$

Lower is better. Penalises large errors more.

MAP@10

Mean Average Precision at 10 — measures ranking quality.

$$\text{AP@10} = (1/\min(10, |\text{rel}|)) * \sum_k P(k) * \text{rel}(k)$$

Relevance threshold: actual rating ≥ 4.0 . Higher is better.

Experimental Results

Model Comparison

Model	RMSE	MAP@10
Global Mean	1.0540	—
User Mean	0.9810	—
Item Mean	0.9825	—
Item-CF	0.8809	—
Funk-SVD ★	0.8151	0.0153

22.6%

RMSE improvement
vs. global mean

0.7897

Best validation
RMSE (epoch 15)

15

Epochs before
early stop

Training Curve (Val RMSE)



Recommendations & Conclusions

Sample — User 305344 (14,123 ratings, eclectic taste)

#1 Elf: Bonus Material (pred: 4.91) **not in test**

#2 In Cold Blood (pred: 4.55) **miss**

#3 The Private Life of Henry VIII (pred: 4.41) **miss**

#4 Best of the Blues Brothers (pred: 4.18) **miss**

#5 Kingdom of Heaven (pred: 4.09) **miss**

Key Takeaways

- Funk-SVD is the strongest model — 22.6% RMSE improvement over baseline, clear MAP@10 advantage.
- Item-CF provides an interpretable complement; real-world systems benefit from blending both approaches.
- Users with eclectic taste challenge MF — a key motivation for content-based and hybrid extensions.
- Future work: Neural CF, temporal dynamics, cold-start metadata, and A/B deployment testing.